

MODULE-5:

This document, the student guide for Module 5 of the AWS Academy Cloud Architecting course, explores the intricacies of **Amazon Elastic Compute Cloud (Amazon EC2)**, a service providing resizable compute capacity in the cloud. The guide explains how to select, configure, and optimize EC2 instances based on various workload requirements.

Understanding the Compute Landscape in AWS

- **AWS Runtime Compute Choices:** The guide presents an overview of the diverse compute options within the AWS ecosystem. It classifies these choices into five categories:
 - **Virtual Machines (VMs):** Represented by Amazon EC2, this category is the focus of the module.
 - **Containers:** Includes Amazon Elastic Container Service (Amazon ECS) and Amazon Elastic Kubernetes Service (Amazon EKS) for containerized applications.
 - **Virtual Private Servers (VPS):** Amazon Lightsail caters to users who need a simplified management interface and predictable pricing.
 - **Platform as a Service (PaaS):** AWS Elastic Beanstalk allows developers to upload code and have the service automatically handle deployment.
 - **Serverless:** AWS Lambda and AWS Fargate enable running code and containers without server management.

Deep Dive into Amazon EC2

- **Amazon EC2:** The document focuses on **Amazon EC2**, a service offering **virtual machines (VMs)** in the cloud. This service enables the provisioning of **EC2 instances**, providing complete control over computing resources. It allows users to:
 - Provision servers in minutes.
 - Scale capacity up or down automatically.
 - Pay only for the capacity used.
- **Virtualization:** EC2 instances run as virtual machines on host computers in AWS Availability Zones. Each VM operates on an OS (such as Amazon Linux or Microsoft Windows), allowing the installation and execution of applications.
 - **Hypervisor Layer:** VMs run on top of a hypervisor layer managed by AWS, responsible for managing VM access to physical hardware resources.
 - **Storage Options:** EC2 instances support various storage options, including:
 - **Instance Store:** Ephemeral storage physically attached to the host computer, providing temporary block-level storage.
 - **Amazon Elastic Block Store (Amazon EBS):** Persistent block storage volumes, persisting data even when the instance is stopped.
 - **EBS-Optimized Instances:** Offer faster access to attached EBS volumes.
- **Use Cases:** The guide discusses several situations where Amazon EC2 is a suitable choice:

- Requirement for complete control over computing resources, including operating system and processor type.
- Desire to optimize compute costs by leveraging various pricing options (On-Demand Instances, Reserved Instances, Spot Instances, Savings Plans, Dedicated Hosts).
- Need to run any type of workload, from simple websites to enterprise and generative AI applications.
- **Provisioning an EC2 Instance:** The document outlines the steps for provisioning an EC2 instance:
 - Choosing an **Amazon Machine Image (AMI)**, the template used to launch an instance.
 - Selecting an **instance type** that defines the configuration of CPU, memory, storage, and networking.
 - Specifying a **key pair** for secure SSH or RDP access.
 - Configuring network placement, addressing, and assigning a public IP or DNS address.
 - Assigning a **security group** (a set of firewall rules) to the instance.
 - Defining storage options (instance store or EBS volumes).
 - Attaching an **IAM role** for API calls to AWS services.
 - Optionally specifying **user data** to automate installations and configurations during launch.

Understanding and Selecting AMIs

- **Amazon Machine Image (AMI):** The document defines an **AMI** as the template for launching an EC2 instance, containing:
 - A template for the root volume, including the operating system and potentially pre-installed software.
 - Launch permissions, controlling which accounts can use the AMI.
 - Block device mappings, specifying any storage volumes to attach to the instance.
- **AMI Benefits:**
 - **Repeatability:** An AMI can be used multiple times to launch identically configured instances.
 - **Reusability:** Instances launched from the same AMI are configured identically, simplifying cluster creation and environment recreation.
 - **Recoverability:** You can create an AMI from a configured instance as a backup, allowing for instance replacement in case of failure.
- **Choosing an AMI:** Several factors influence AMI selection:
 - **Region:** AMIs are region-specific, so choose one in the target region for the instance.
 - **Operating system:** Options include Microsoft Windows and various Linux distributions.
 - **Storage for the root device:** AMIs can be either **Amazon EBS-backed** (persistent storage) or **instance store-backed** (temporary storage).
 - **Architecture:** Select the appropriate processor architecture (32-bit/64-bit, x86/ARM).

- **Virtualization type:** Use an AMI with **Hardware Virtual Machine (HVM)** virtualization for optimal performance.
- **AMI Sources:**
 - **Quick Starts:** AMIs provided by AWS for Linux and Microsoft Windows.
 - **My AMIs:** User-created AMIs.
 - **AWS Marketplace:** Pre-configured templates from third-party vendors.
 - **Community AMIs:** Publicly shared AMIs (use with caution).
- **Instance Store-Backed vs. Amazon EBS-Backed AMI:** The guide highlights key differences between these AMI types:
 - **Boot time:** EBS-backed instances boot faster.
 - **Maximum root device size:** EBS-backed instances support a larger root device.
 - **Ability to stop the instance:** Only EBS-backed instances can be stopped.
 - **Instance type change:** You can change the instance type of an EBS-backed instance by stopping it.
 - **Charges:** EBS-backed instances incur EBS volume and snapshot storage costs, while instance store-backed instances incur Amazon S3 storage costs.
 - **Use case:** EBS-backed for persistent storage, instance store-backed for temporary storage.
- **Amazon EC2 Instance Lifecycle:** The document describes the different states an EC2 instance goes through:
 - **Pending:** Instance is being provisioned and booting.
 - **Running:** Instance is fully booted and ready for use.
 - **Rebooting:** Instance is being rebooted, retaining data on instance store volumes.
 - **Terminating:** Instance is being shut down permanently.
 - **Stopping:** EBS-backed instances can be stopped, saving costs.
 - **Hibernating:** EBS-backed instances can be hibernated, saving in-memory storage, private IP, and Elastic IP.
- **Creating a New AMI:** The guide provides a step-by-step explanation of creating an AMI from an existing EC2 instance.
- **EC2 Image Builder:** The guide introduces **EC2 Image Builder**, a service that automates the creation, management, and deployment of updated and compliant golden VM images.

Understanding and Selecting Instance Types

- **EC2 Instance Type Configuration:** An **instance type** defines the CPU, memory, storage, and network performance characteristics of an instance. The choice depends on workload requirements and cost constraints.
- **EC2 Instance Type Name:** The document breaks down the naming convention for EC2 instance types, which includes information about the family, generation, processor family, additional capabilities, and size.
- **Suitability of Instance Types for Workloads:** The guide categorizes instance types based on their suitability for different workloads:
 - **General purpose:** Offer a balance of resources for diverse workloads.
 - **Compute optimized:** Suitable for compute-bound applications.
 - **Storage optimized:** Designed for workloads requiring high sequential read/write access to large datasets.

- **Memory optimized:** Offer fast performance for workloads processing large datasets in memory.
- **Accelerated computing:** Use hardware accelerators for tasks like floating-point calculations and graphics processing.
- **High performance computing (HPC) optimized:** Purpose-built for running HPC workloads at scale on AWS.

Choosing an Instance Type and AWS Compute Optimizer

- **Choosing an Instance Type:** The guide acknowledges the challenge of selecting the optimal instance type from the extensive selection available. It advises considering performance needs, cost requirements, and using the **Instance Types page** in the EC2 console for comparison and filtering.
- **AWS Compute Optimizer:** The document introduces **AWS Compute Optimizer**, a service that analyzes EC2 instances and Auto Scaling groups, providing recommendations to reduce cost and improve performance.

Understanding Storage Options for EC2

- **Amazon EC2 Storage Overview:** The document provides an overview of the different storage options available for EC2 instances:
 - **Instance store:** Temporary block-level storage physically attached to the host.
 - **Amazon EBS:** Persistent block storage volumes that can be attached to instances in the same Availability Zone.
 - **Amazon EFS:** Shared file system for Linux instances.
 - **Amazon FSx for Windows File Server:** Shared file system for Microsoft Windows instances.
- **Amazon EBS SSD-Backed Volume Types:** The guide describes the two types of SSD-backed volumes:
 - **General Purpose SSD (gp2):** Balanced performance for various workloads.
 - **Provisioned IOPS SSD (io1):** Highest performance option, suitable for mission-critical, low-latency workloads.
- **Amazon EBS HDD-Backed Volume Types:**
 - **Throughput Optimized HDD (st1):** Low-cost option for frequently accessed, throughput-intensive workloads.
 - **Cold HDD (sc1):** Lowest-cost option for less frequently accessed workloads.
- **Amazon EBS-Optimized Instances:** Certain instance types offer dedicated network connections to EBS volumes, increasing I/O performance.
- **Shared File Systems:** The document addresses the need for sharing data among multiple EC2 instances. It introduces:
 - **Amazon Elastic File System (Amazon EFS):** Provides a scalable and elastic file system for Linux instances.
 - **Use Cases:**
 - Home directories
 - Enterprise application file systems
 - Application testing and development

- Database backups
- Web serving and content management
- Media workflows
- Big data analytics
- **Amazon FSx for Windows File Server:** Offers fully managed Windows file servers for Microsoft Windows instances.
 - **Use Cases:**
 - Home directories
 - Lift-and-shift applications
 - Media and entertainment workflows
 - Data analytics
 - Web serving and content management
 - Software development environments

Other EC2 Configuration Considerations

- **EC2 Instance User Data:** **User data** allows the execution of initialization scripts (shell scripts or cloud-init directives) during instance launch. This enables tasks like:
 - Software updates and patching
 - License key retrieval and installation
 - Installation of additional software
- **Retrieving Instance Metadata:** The document explains how to retrieve **instance metadata**, providing information about the running instance, through a specific URL.
- **Working with User Data on Running Instances:** The guide outlines steps for modifying the user data script on a running instance.
- **Manually Running Commands Best Practices:** The document recommends updating the user data script even when commands are run manually on the instance to maintain a configuration record and for reusability.
- **AMI Deployment Models:** The guide introduces three main AMI deployment models:
 - **Basic AMIs:** Base AMIs with OS-only configurations, offering full configurability and upgradeability.
 - **Silver AMIs:** Partially pre-configured AMIs, providing a balance between boot speed and build time.
 - **Golden AMIs:** Fully configured and customized AMIs, ensuring consistent behavior across instances.
- **Placement Groups:** **Placement groups** allow control over the placement of interdependent instances within an Availability Zone, offering benefits like increased network performance and reduced correlated failures. The guide describes the different placement strategies:
 - **Cluster:** Packs instances close together.
 - **Partition:** Spreads instances across logical partitions.
 - **Spread:** Places a small group of instances across distinct hardware.

Amazon EC2 Pricing Models and Strategies

- **AWS Free Tier:** New AWS customers can access limited EC2 resources for free for 12 months.
- **Amazon EC2 Purchase Models:** The document outlines different purchase models for EC2 instances:
 - **On-Demand Instances:** Provide flexibility with no long-term commitments.
 - **Reserved Instances:** Offer significant discounts for 1-year or 3-year commitments.
 - **Savings Plans:** Offer discounts similar to Reserved Instances with greater flexibility.
 - **Amazon EC2 Spot Instances:** Provide spare capacity at substantial savings.
- **Amazon EC2 Capacity Reservations:** Allow reserving compute capacity in a specific Availability Zone for guaranteed access. The guide describes:
 - **On-Demand Capacity Reservations:** Guarantee access to capacity for any duration.
 - **Amazon EC2 Capacity Blocks for ML:** Enable reserving GPU instances for future machine learning workloads.
- **Amazon EC2 Dedicated Options:** These options provide instance capacity on physically dedicated servers:
 - **Dedicated Instances:** Provide per-instance billing and automatic placement on isolated hosts.
 - **Dedicated Hosts:** Offer per-host billing, visibility into sockets and cores, and control over instance placement.
- **Amazon EC2 Cost Optimization Guideline:** The guide recommends combining various purchase options to optimize costs. The general approach is to:
 - Run steady-state workloads on Reserved Instances or Savings Plans.
 - Use On-Demand Instances for spiky workloads.
 - Utilize Spot Instances for fault-tolerant, stateless workloads.

Applying the AWS Well-Architected Framework to Compute

- The document emphasizes aligning EC2 deployments with the **AWS Well-Architected Framework** to ensure secure, efficient, and cost-optimized solutions. It highlights relevant best practices from the framework's pillars:
 - **Security:** Automate compute protection by leveraging tools like EC2 Image Builder and user data scripts.
 - Control traffic at all layers using security groups and network segmentation.
 - **Performance Efficiency:** Select the most appropriate compute options (instance types and storage types) based on workload requirements.
 - **Cost Optimization:** Right-size compute resources, choose the best pricing model (On-Demand, Reserved, Savings Plans, Spot), and leverage AWS tools for analysis and recommendations.
 - **Sustainability:**
 - Use the minimum amount of hardware by right-sizing resources.
 - Select instance types with the least environmental impact.

- Utilize managed services to benefit from AWS's sustainability optimizations.

This detailed breakdown encompasses the essential points from Module 5 of the AWS Academy Cloud Architecting course, providing an in-depth understanding of Amazon EC2 and its various facets. Please refer to the original document for more detailed explanations, examples, and hands-on exercises.